

Claudio Camolese

Curriculum Vitae

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EDUCATION

ENS Paris-Saclay - Université Paris-Saclay, France

2026-2027

Master of research in Applied Mathematics, MVA track - Mathematics, Vision and Learning

Highly selective programme (typically best 5–10%) taught by leading researchers as Cordelia Schmid, Jean Ponce, and Ivan Laptev.

Politecnico di Torino, Italy

École Nationale Supérieure d'Informatique et de Mathématiques Appliquées, France

2024-2026

Double Master's degree program, specializing in Artificial Intelligence.

Relevant courses: *Robotics, Robot learning, Computer Vision, Deep Natural Language Processing, GPU computing, Generative multimodal AI, Machine learning and Deep learning, Mathematics in machine learning*

Politecnico di Torino, Italy

2021-2024

Bachelor's degree in Physics Engineering.

Relevant courses: *Advanced Methods for Physics, Quantum physics, Solid state physics, Advanced electromagnetics*

AI & ROBOTICS EXPERIENCE

University of Cambridge and TUM

October 2026 – February 2027

Research Intern – Supervisors: Prof. Wysocki, Prof. Wang

Incoming Research Intern on *Interactive World-Action Models for Open-World Robotic Manipulation*.

Politecnico di Torino, VANDAL team

September 2026 – present

Research Intern – Supervisors: Prof. Averta, PhD Buoso

Research Intern on *In-Context Learning and Zero-Shot Generalization for Robotic Manipulation*.

University of Cambridge, CV4DT team

February 2026 – Present

Research Intern – Supervisors: Prof. Wysocki, Prof. Wang, Prof. Averta, Prof. Pineda

Physics-Informed Safety: Developed a plug-and-play safety layer for Vision-Language-Action models that applies minimal corrections to unsafe actions while preserving the intended task.

Vision-Language Perception: Integrated Qwen2.5-VL and SAM3 to identify task-relevant objects and exclude manipulation targets and placement goals from the obstacle representation.

SafeLIBERO Evaluation and Metric Design: Evaluated PhySVLA on 1,200 SafeLIBERO episodes per configuration across three suites and two safety levels, using Safe Success Rate (SSR) metric. With frozen $\pi_{0.5}$, PhySVLA improves TSR from 62.7% to 70.9%, CAR from 18.2% to 77.2%, and SSR from 16.7% to 61.0%; with OpenVLA-OFT, it improves TSR from 28.2% to 32.1%, CAR from 13.4% to 55.9%, and SSR from 6.9% to 26.2%, demonstrating transfer across distinct VLA backbones.

Politecnico di Torino, Squadra Corse Driverless

September 2024 – October 2025

Perception and AI engineer - Software team member

LiDAR-Based State Estimation: LiDAR-Based State Estimation: Developed components of the vehicle state-estimation pipeline with an Ouster OS0 sensor, leveraging ROS2, Eigen, Sophus, oneTBB, and Open3D.

Reinforcement Learning Integration: Integrated a reinforcement-learning agent into MATLAB and Simulink for operation on pre-mapped tracks, enabling the vehicle to exploit known track geometry and reducing its reliance on online perception.

Formula SAE Driverless: Supported the development and testing of the autonomous racing system and competed in Formula SAE Driverless competitions.

Memory-Efficient LLM Inference: Investigated inference strategies for deploying reasoning Large Language Models on memory-constrained edge devices using DeepSeek-R1.

Recursive Reasoning Pipeline: Designed a recursive inference pipeline that iteratively feeds intermediate outputs back into the model, decomposing the reasoning process into sequential steps that remain within the device memory constraints.

AI & ROBOTICS PROJECTS

Temporal Mixture-of-Experts for Quadruped Parkour Locomotion: Extended an IsaacLab-based perceptive locomotion framework for the Unitree Go2 by introducing a Temporal Mixture-of-Experts (MoE) policy architecture for quadruped parkour. Developed and evaluated teacher–student locomotion policies, using temporal expert routing to improve adaptation across heterogeneous terrain and parkour behaviours.

RGB-D 6D Object Pose Estimation: Developed an end-to-end pipeline for estimating object position and orientation from RGB-D observations. Evaluated the approach on the LINEMOD dataset using the ADD(-S) metric, and adapted it for real-time traffic-cone pose estimation in the Squadra Corse Driverless simulator while the vehicle was in motion.

Semantic Scene Understanding for Autonomous Drones: Implemented a DeepLabV3+ architecture for semantic segmentation of aerial imagery, enabling pixel-level scene understanding for autonomous navigation and environmental perception.

Sim-to-Real Reinforcement Learning: Developed robust control policies for the MuJoCo Hopper under source–target dynamics shifts. Applied Uniform Domain Randomization to robot link masses and evaluated its effectiveness in reducing performance degradation when transferring policies across different dynamics.

Generative Models for Image Synthesis: Designed and implemented diffusion models, flow-matching models, GANs, and VAEs from scratch, including their architectures, training objectives, sampling procedures, and optimization pipelines. Compared the different generative paradigms in terms of training stability, sample quality, and inference behaviour.

AWARDS & ACHIEVEMENTS

UROP@PoliTO Open Badge: Awarded by Politecnico di Torino upon completion of a supervised Undergraduate Research Opportunities Programme project on memory efficient LLM inference for edge devices.

ManagerItalia Academic Excellence Scholarships: Recipient of five merit-based scholarships awarded by Manageritalia and the Fondo Mario Negri in recognition of sustained academic achievement.

Stellantis Student Award: Recipient of a Stellantis merit-based student award for academic excellence.

Imprenditori #GenNext: Selected for the entrepreneurship programme organised by UniCredit and the Club degli Investitori, focused on start-up creation, fundraising, and intellectual property.

SKILLS

Programming: Python, C++, CUDA, MATLAB, Java

AI & Robotics Tools: PyTorch, LIBERO, MuJoCo, PyBullet, ROS 2, Gazebo, RViz, OpenCV, LiDAR

LANGUAGES

Italian: Native speaker

English: IELTS C1

French: DELF B1